

Power Function and Uniformly Most Powerful Test

Power function

$$H_0: \theta \in \Theta_0 \text{ vs } H_a: \theta \in \Theta_0^c$$

$$P_\theta(\mathbf{X} \in R) = \begin{cases} \text{probability of a Type I Error,} & \text{if } \theta \in \Theta_0 \\ \text{one minus the probability of a Type II Error} & \text{if } \theta \in \Theta_0^c. \end{cases}$$

The power function of a hypothesis test with rejection region R is the function of θ defined by $\pi(\theta) = P_\theta(\mathbf{X} \in R)$.

$$\pi(\theta) = \begin{cases} 0 & \theta \in \Theta_0 \\ 1 & \theta \in \Theta_0^c \end{cases}$$

Ideal power function



Power function: Example

Let $X \sim \text{bin}(5, \theta)$. Consider testing $H_0 : \theta \leq \frac{1}{2}$ versus $H_a : \theta \geq \frac{1}{2}$

First test : that rejects H_0 if and only if all “successes” are observed.

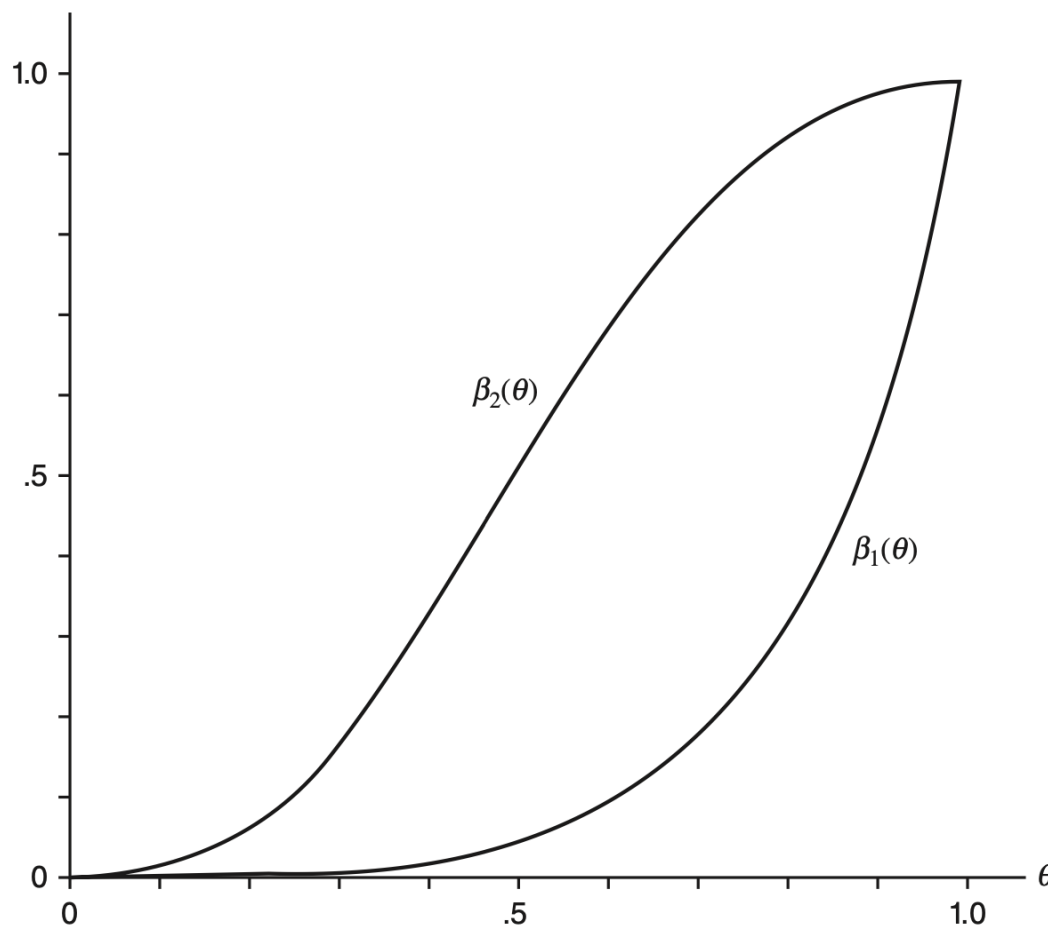
$$\pi_1(\theta) = \theta^5$$

Second Test : reject H_0 if $X \geq 3$

$$\pi_2(\theta) = \binom{5}{3} \theta^3(1-\theta)^2 + \binom{5}{4} \theta^4(1-\theta)^1 + \binom{5}{5} \theta^5(1-\theta)^0.$$

Power function: Example

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Read: β as π

Ack: Casella, Berger

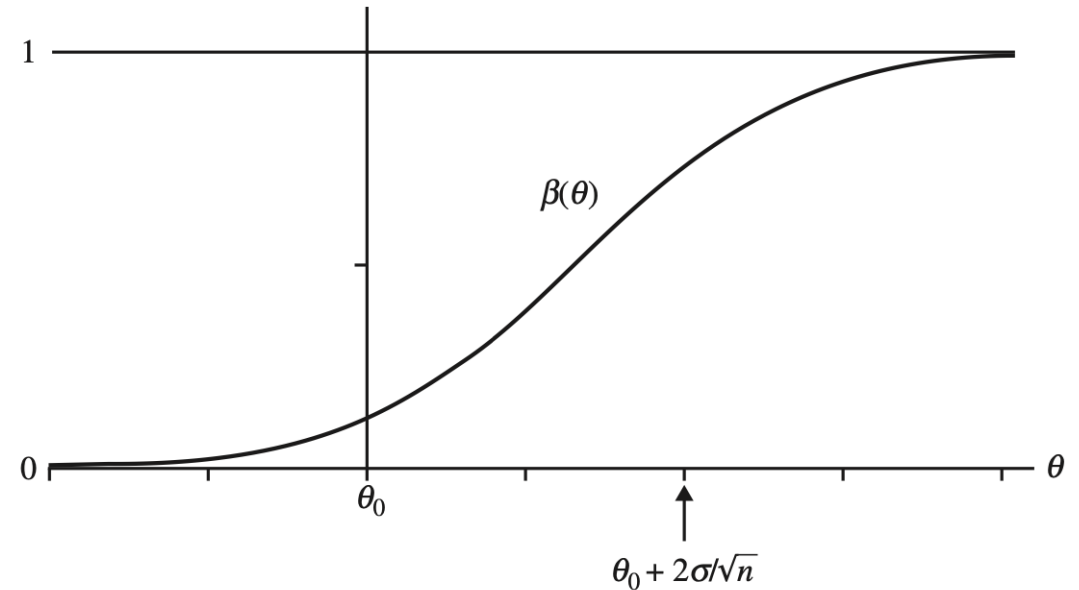
Power function: Example

$$H_0: \theta \leq \theta_0 \quad \text{vs} \quad H_a: \theta_1 \geq \theta_0$$

Assumptions: Normal population with known variance σ

Rejects H_0 if $\frac{\bar{X} - \theta_0}{\sigma/\sqrt{n}} > c$

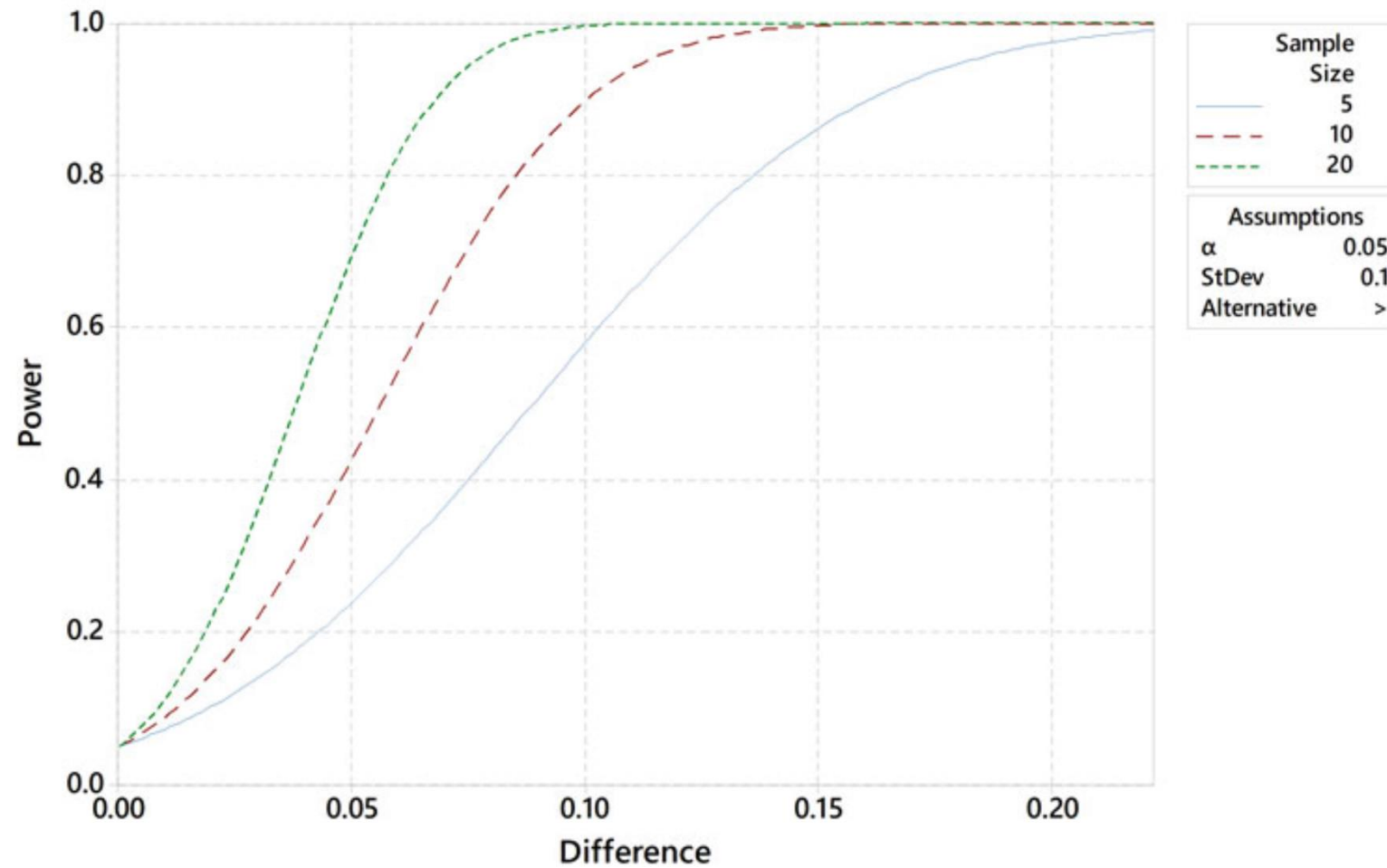
$$\begin{aligned}\pi(\theta) &= P_\theta \left(\frac{\bar{X} - \theta_0}{\sigma/\sqrt{n}} > c \right) \\ &= P_\theta \left(\frac{\bar{X} - \theta}{\sigma/\sqrt{n}} > c + \frac{\theta_0 - \theta}{\sigma/\sqrt{n}} \right) \\ &= P \left(Z > c + \frac{\theta_0 - \theta}{\sigma/\sqrt{n}} \right),\end{aligned}$$



Read: β as π

Ack: Casella, Berger

Power function: Dependence on sample size



Ack: Devore et. al

Power function: Dependence on sample size

$$H_0 : \theta \leq \theta_0 \quad \text{vs} \quad H_a : \theta \geq \theta_0$$

Assumptions : Normal population with known variance σ

Test procedure : Rejects H_0 if $\frac{\bar{X} - \theta_0}{\sigma/\sqrt{n}} > c$. Estimate the size of the sample to be chosen so that Type I error is 0.1 and Type II error is 0.2 if $\theta \geq \theta_0 + \sigma$.

$$\pi(\theta_0) = 0.1 \qquad \pi(\theta_0 + \sigma) = 0.2$$

$$P(Z \geq c) = 0.1 \Rightarrow c = 1.28 \quad P(Z > c - \sqrt{n}) = 0.8 \Rightarrow c - \sqrt{n} = -0.84 \Rightarrow n = 4.49$$

Choose $n = 5$ (why?)

Neyman – Pearson Lemma : Application

$$H_0 : \mu_0 \quad H_a : \mu_1 ; \mu_1 > \mu_0$$

Assumptions: Normal population with known variance σ

Likelihood function:

$$L = \prod_{i=1}^n \frac{\exp \left[-\frac{1}{2} \left(\frac{x_i - \mu}{\sigma} \right)^2 \right]}{\sqrt{2\pi}\sigma} = \frac{\exp \left[-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 \right]}{(2\pi\sigma^2)^{n/2}}.$$

Simplification of $\frac{L_1}{L_0} \geq k$ gives an equivalent and simplified definition of critical region

Ack: Hoel, Port, Stone

Neyman – Pearson Lemma : Application

Simplification of $\frac{L_1}{L_0} \geq k$ gives an equivalent and simplified definition of critical region

Since $\mu_1 > \mu_0$:

$$\sum x_i \geq \frac{\sigma^2 \log k}{\mu_1 - \mu_0} + \frac{n(\mu_1 + \mu_0)}{2} \dots\dots\dots (1)$$

$$\forall c \in (0, \infty) \text{ s.t. } \sum x_i \geq c \Rightarrow \exists k \in (0, \infty)$$

Hence any $c \in (0, \infty)$ gives a valid test

Ack: Hoel, Port, Stone

Neyman – Pearson Lemma : Application

$$c = \mu_0 + \Phi^{-1}(1 - \alpha) \frac{\sigma}{\sqrt{n}}$$

Depends on μ_0 and α



Thus for any $\mu_1 > \mu_0$ we get an optimal test according to NP lemma

UPM

Let $\mathcal{C}$ be a class of test for testing $H_0: \theta \in \Theta_0$ vs $H_a: \theta \in \Theta_0^c$. A test in class $\mathcal{C}$, with power function $\pi(\theta)$, is *uniformly most powerful* (UMP) class $\mathcal{C}$ test if $\pi(\theta) \geq \pi'(\theta)$ for every $\theta \in \Theta_0^c$ and every $\pi'(\theta)$ that is a power function of some test in class $\mathcal{C}$.